

AyuRAG: Ayurvedic Knowledge Integration with Retrieval-Augmented Lightweight Large Language Models

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Abstract— Ayurveda, a traditional system of medicine, is boundless and rich in historical connotations, but it often remains unutilized because of fragmented data sources as well as the absence of access to properly structured, substantial literature. Healthcare professionals face difficulties in accessing and integrating Ayurvedic knowledge, hindering their ability to derive actionable understandings from comprehensive literature. Traditional methods are not adequate to provide structured, accessible databases, which limits the utility of Ayurveda in modern healthcare. To bridge this gap, a Large Language Model (LLM) with Retrieval-Augmented Generation (RAG) is proposed in this work. This system enables concise, context-dependent retrieval of Ayurvedic knowledge and uses RAG to dynamically retrieve relevant chunks of information from the repository based on the query and uses LLMs to generate contextually accurate responses. This hybrid response accounts for highly specific and relevant responses. By utilizing Ayurvedic data from unstructured, publicly available sources, this system is structured around key medical domains, focusing on diseases and their cures as per Ayurveda. Additionally, various techniques were used for faster inference and mapping for more accurate and efficient workflow. This approach overcomes shortcomings like fragmentation and inaccessibility of Ayurvedic data. The scalability of this approach provides future expansion, offering a versatile platform for integrating Ayurvedic principles into modern medical practices. In future expansions a wider dataset with a focus on other aspects of Ayurveda could be included, extending the capabilities to turn it into a more specialized Ayurvedic care for individuals.

Keywords— Ayurveda; Large Language Models (LLMs), LoRA, Fine Tuning, Retrieval-Augmented Generation (RAG), Vector database

I. INTRODUCTION

Ayurveda, a system of natural healing, that dates back to ancient times, has a legacy. The traces back over 5000 years to the Vedic civilization in India [1]. “Ayur” meaning life and “Veda” meaning science and knowledge, mixes the meaning of Ayurveda, “The science of life.” Ayurveda is far more than a system of medicine; it is a comprehensive way of life that prominences the balance of not only the body but mind and

spirit too [2]. The Ayurvedic way of life focuses attention on the preservation of equilibrium among three of the crucial doshas: Vata, Pitta, and Kapha, focused on the achievement of wholesome health. This preservation of balance is central to Ayurveda, where health is defined as a state of equilibrium, not only of the physical body but also of the spiritual states and mental states. Ayurveda’s origin is found to be deeply rooted in the ancient Indian Vedic texts, and these are said to be the oldest written records [3], [4].

Ayurvedic knowledge, while vast and rich, is scattered across various fragmented formats including text documents, images, PDF files, and presentations from diverse sources such as academic journals, traditional manuscripts, and digital archives. These data sources, while invaluable, often exist in an unstructured form or different languages such as Hindi and Sanskrit, lacking uniformity in terminology and presentation [5]. This lack of standardization, combined with the wide range of formats, creates significant challenges for healthcare practitioners and researchers who require quick and reliable access to Ayurvedic knowledge [6-8].

To bridge this gap, a framework utilizing Retrieval-Augmented Generation (RAG) combined with large language models (LLMs) has been proposed. The RAG approach blends two core components: a retriever that efficiently searches through structured Ayurvedic knowledge bases and a generator that crafts responses grounded in the retrieved content [9], [10]. This synergy ensures that responses are accurate, relevant, and context-sensitive, tackling common issues like misinformation and hallucinations often found in generative AI models.

To align the model closely with Ayurvedic principles, domain-specific datasets have been utilized which were sourced from reputable government websites and related datasets. This ensures that the output remains faithful to Ayurvedic guidelines, enhancing both reliability and trustworthiness. Additionally, real-time processing capabilities provide personalized responses, allowing the model to serve as a bridge between traditional medical wisdom and modern AI-driven solutions [11]. To support fine-tuning for domain-specific adaptation, the proposed methodology incorporates parameter-

efficient techniques that enable streamlined learning of Ayurvedic knowledge without significant resource demands. This balance maintains the general linguistic capabilities of the model while embedding specialized content, facilitating smooth deployment across various platforms and ensuring responsive interactions.

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The proposed approach, as shown in Fig.1, addresses these challenges by implementing an RAG architecture fine-tuned with Ayurveda-specific knowledge. This strategy not only enhances the model's capacity for precise information retrieval but also supports real-time, context-aware response generation. The result is a robust system designed to integrate seamlessly with healthcare practices, overcoming traditional limitations and making Ayurvedic knowledge more accessible and practical.

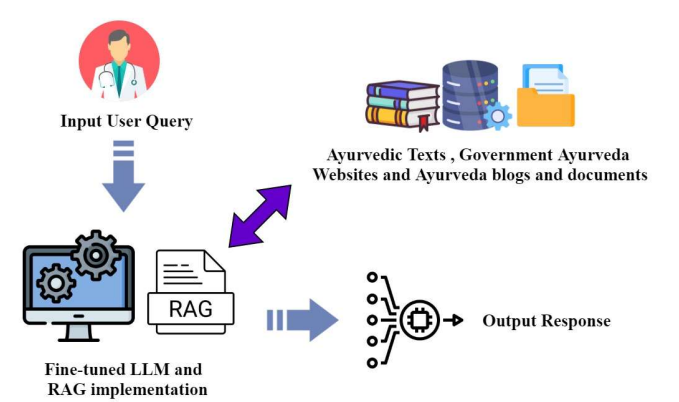


Fig. 1. AI-Powered Retrieval-Augmented Generation Framework with Large Language Models for Enhanced Ayurveda Response

The main contributions of the proposed work are as follows:

- (a) Developed a Retrieval-Augmented Generation (RAG) model trained on Ayurvedic knowledge derived from reliable sources such as the Ministry of AYUSH, Government of India, so that the advice given will not only be contextually correct but also medically sound in the sense that it is Ayurveda specific.
- (b) By utilizing methods for efficient model adaptation, the proposed system incorporates Ayurvedic terminology and principles into the LLM, minimizing hallucinations and

- enhancing the semantic understanding of Ayurvedic data, enhancing greater trust between healthcare providers and patients.
- (c) Integrated techniques for low-latency AI inference, allowing real-time, contextually relevant responses, optimizing both retrieval and generation stages for responsive and accurate Ayurvedic recommendations.
- (d) The architecture is designed for scalability and efficiency, providing a balanced solution that addresses the inherent limitations of AI in healthcare by improving information retrieval, reducing computational complexity, and offering reliable, structured Ayurvedic comprehension.

The paper organization is given as follows: The details about the dataset and proposed work are given in Section II. The experimental results are discussed in Section III. Conclusions and future works associated with the proposed work are provided in Section IV.

II. PROPOSED WORK AND METHODOLOGY

A. Dataset Description

The LLM with domain-specific datasets has been trained, including the texts from the Ministry of AYUSH and Kaggle Question-Answer datasets, to be sure that the model is aligned with Ayurvedic principles [17]. The Ayurveda LLM model will be based on three main datasets, each serving a specific purpose in the development and evaluation of this question-answering system:

- (a) *JSON Question-Answer Pairs*: This contains a total of 7,382 rows, split between 70% training set and 30% test. These pairs are quite important to the training of the model for understanding and responding appropriately to Ayurveda-related queries.
- (b) *TXT Data for RAG (Retrieval-Augmented Generation)*: These are the text files with the related text documents acting as a knowledge base for the RAG system. Ministry of Ayush's data consisted mainly of blogs, official documents, and presentations in different formats like .xlsx, .pdf, and .ppt. All these were cleaned up and preprocessed to get uniformity in output text compatible with the RAG system.
- (c) *CSV Data for Model Evaluation*: It is a set of supplementary question-answer pairs curated specifically to test the model. This dataset would serve as a validation set that could measure the accuracy and efficiency of an Ayurveda LLM + RAG model in authentic scenarios. Dataset components are summarized in Table 1.

TABLE I. COMPLETE DATASET DESCRIPTION AND ITS CHARACTERISTICS

Data Subset	Description	Dataset Component	Usage %
JSON Q&A Pairs	5,168 rows for fine-tuning	Training Set	70%
JSON Q&A Pairs	2,214 rows for fine-tuning	Test Set	30%
TXT Data	Processed Ayurveda resources from the Ministry of Ayush	RAG Knowledge Base	100%
CSV Data	Q&A pairs for evaluation	Validation Set	100%

This comprehensive dataset ensures that the model is well-equipped to provide answers to a vast range of Ayurveda-related queries. The RAG module provides fine-tuned question-answering capabilities in combination with this huge knowledge base that enables the model to provide both concise answers as well as providing in-depth explanations when needed.

B. Proposed Methodology

The diagram to explain the complete methodology in the form of various blocks in the sequence is shown in Fig. 2. In the proposed framework, the first step is that of gathering both structured and unstructured data from different sources. The main dataset is the documents retrieved from the Ministry of AYUSH, which reflect complete information about Ayurveda and further elaborate on diseases and traditional treatment for them. The availability of the documents varies with respect to their formats in terms of PDF, Excel spreadsheets, and PowerPoint slides, which makes it full and diverse knowledge.

As outlined in Fig.3, methodology begins with the extraction of data from those knowledge sources, as diagrammed below. The extracted data is then fed through text splitting: the splitting of documents into smaller, more manageable pieces. All these chunked documents, along with a unique set of question-and-answer pairs specially prepared, make up the input for us to train and fine-tune the proposed model. These blocks of text and Q&A pairs are further processed by the tokenizer to make suitable tokens for the proposed language model. An embedding model transforms these tokens into vector representations thus capturing the semantic nuances of Ayurvedic concepts. The vectors are then stored in ChromaDB, the chosen vector database, which stores and retrieves the embedded information efficiently.

Llama-3-8B-Instruct has been selected as the base model, onto which the Ayurvedic dataset has been fine-tuned [18], [19]. The proposed model is also quantized for better performance and easier deployment at 4-bit precision. It leads to a drastic model size reduction without loss of effectiveness so that inference and deployment can be done easily and very fast. unsloth library is an important part of the process, making fine-tuning and inference much faster. It accelerates and makes fast and reliable development iterations in continually refining the model's understanding of Ayurvedic concepts.

In the same way, when a user submits a query, the tokenization and embedding process occurs. Then, the system uses the retriever to do a similarity or semantic search in the vector database to determine the most relevant documents to the query within the top N ranks. The final retrieved documents along with the actual query from the user serve as the context in which the trained LLM generates its response. BLEU, ROUGE, and F1 score, as mentioned in section III, are performance metrics to measure the quality, relevance, and accuracy of the responses generated for Ayurvedic knowledge in the given context. Then parameters will be fine-tuned such as top-p, temperature, max new tokens, and repetition penalty to optimize its performance.

The proposed approach will provide a light, highly deployable model which carries with it several benefits in integrative Ayurvedic knowledge. It will make information

about Ayurveda complex and accessible to practitioners and laymen, thus being able to bridge the chasm of traditional wisdom to modern users. The model allows for standardized information derived from a set of authoritative sources, which would thus ensure consistency with any knowledge being promulgated about Ayurveda. The scalability of architecture enables processing a vast number of queries at one time, thereby making Ayurvedic information highly accessible to many.

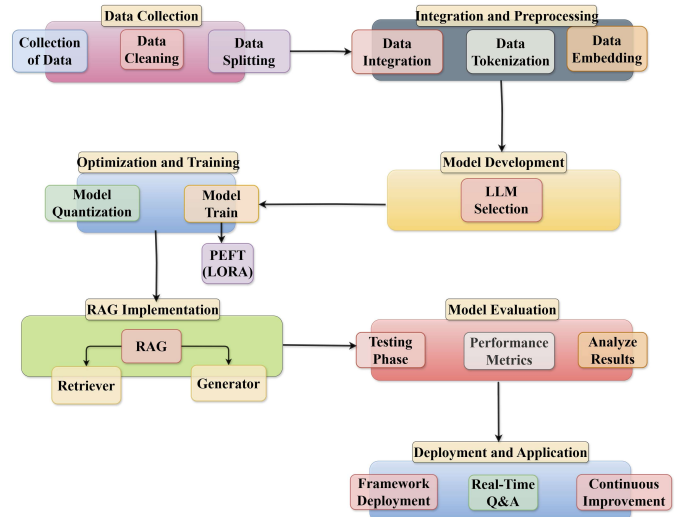


Fig. 2. Proposed development and implementation of the question-answering system using LLMs and RAG techniques

In addition to the improvement in flexibility, the RAG architecture allows straightforward updating of the knowledge base without the need for retraining a complete model to ensure that the information remains current and relevant. 4-bit quantization, optimized architecture ensures fast response times and reduces resource requirements, hence, operating and maintaining it proves efficient and cost-effective. This system of building a robust, efficient, and specific Ayurveda question-answering system combines the best of large language models with a specifically curated set of Ayurvedic knowledge-possibly what makes this traditional Indian medical knowledge more accessible and applicable in a modern context.

III. EXPERIMENTAL RESULTS AND DISCUSSION

The study was based on the integration of LLM with RAG techniques to build a question-answering system around Ayurveda. The experiments were performed on a Nvidia A100 GPU server carrying 40 GB HBM2 memory and 6912 CUDA cores. The framework was Python-driven and used a combination of libraries such as PyTorch, Transformers, and SentenceTransformers.

Four different configurations have been tested, with variations in the following parameters.:

- max new tokens:** maximum tokens that the model is allowed to generate for its response.
- Temperature:** This is the last mechanism: Temperature introduces randomness in selecting the tokens of the model. Higher temperatures, such as 0.8, generate more

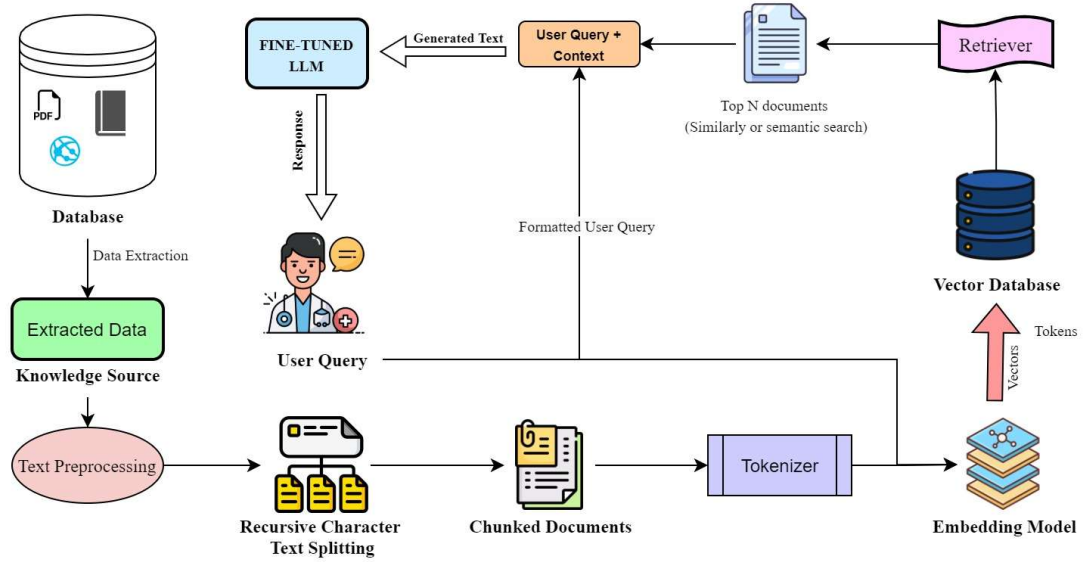


Fig. 3. Proposed Retrieval-Augmented Generation Framework with Large Language Models for Enhanced Ayurveda Response

diverse and creative outputs while lower temperatures, such as 0.2, result in more deterministic and focused output. A moderate temperature of 0.6 has been used that balances creativity and coherence

- (c) **repetition_penalty**: This parameter discourages the model from repeating the same phrases or sentences. A value greater than 1.0 applies a penalty to previously generated tokens, reducing repetition. The experiments included values like 1.2 and 1.05, aiming to maintain coherent yet diverse responses.
- (d) **top_p (nucleus sampling)**: Sets a probability threshold for sampling tokens. The model reads in the most likely tokens whose cumulative probability exceeds that threshold. A larger value, like 0.95 will yield more diverse outputs while a smaller value, like 0.5, constrains it to more probable and therefore possibly more pertinent tokens. The model has been run with values of 0.9 and 0.95 to balance out diversity and relevance in the answers. All the hyperparameters are given in Table II with respective ranges of experimentations and obtained optimum values.

TABLE II. PARAMETERS FOR EXPERIMENTATION AND OPTIMAL VALUES

Parameter	Role/Importance	Optimal Value	Possible Range of Values
max_new_tokens	Controls the length of generated responses	100	[80, 100, 150, 200, 512]
Temperature	Influences randomness and diversity of generated text	0.6	[0, 10e-6, 10e-2, 0.1, 1.2, 1.5, 2]
repetition_penalty	Reduces redundancy in responses	1.2	1.0- 1.5
top_p	Controls diversity of token selection	0.9	0.3-1.0

The proposed model has been evaluated using several metrics, each providing a unique aspect of the model's performance:

(a) **BLEU (Bilingual Evaluation Understudy)**: BLEU is a precision-based metric that measures the overlap of n-grams between the generated text and reference texts. Calculated as given in Eqn. (1):

$$BLEU = BP \cdot \exp \left(\sum_{n=1}^N w_n \log p_n \right) \quad (1)$$

Here, BP refers to the brevity penalty; w_n is the weights of a system with respect to different n-gram levels; and p_n denotes the n-gram precisions. Between 0 and 1, higher BLEU scores mean better performance. Designed mainly for evaluating the quality of machine translation tasks, BLEU also applies to a broader range of text generation tasks.

(b) **ROUGE-1 (Recall-Oriented Understudy for Gisting Evaluation)**: ROUGE-1 measures the overlap of unigrams, that is, single words, between the generated text and reference texts. It provides three sub-metrics, as provided in Eqns. (2-5)

$$\text{Precision} = \frac{\text{number of overlapping unigrams}}{\text{total unigrams in the generated text}} \quad (2)$$

$$\text{Recall} = \frac{\text{number of overlapping unigrams}}{\text{total unigrams in the reference text}} \quad (3)$$

$$\text{F1 score} = \frac{2 \times \text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}} \quad (4)$$

ROUGE-1 is particularly useful for assessing the content overlap at the word level.

(c) **ROUGE-2**: Unlike ROUGE-1, ROUGE-2 calculates the overlapping of bigrams, in which two words form a bigram. The same formulas were applied as those of ROUGE-1 but for the bigrams. ROUGE-2, more sensitive to the word order, could pick up fluency and grammatical correctness much better.

(d) ROUGE-L: ROUGE-L measures the longest common subsequence (LCS) between the generated and reference texts. It computes as given in Eqns. (5-7):

$$R_{lcs} = \frac{LCS(X,Y)}{m} \quad (5)$$

$$P_{lcs} = \frac{LCS(X,Y)}{n} \quad (6)$$

$$F_{lcs} = \frac{(1+\beta^2)R_{lcs}P_{lcs}}{R_{lcs}+\beta^2P_{lcs}} \quad (7)$$

where $LCS(X, Y)$ represents the Longest Common Subsequence of X and Y , respectively. R_{lcs} and P_{lcs} represent LCS-based recall and precision, and β is conventionally set to favour recall such that $\beta > 1$. ROUGE-L is important when it is necessary to capture sentence-level structure similarities. This set of ROUGE metrics provides a single, all-encompassing evaluation of the quality of text produced concerning these various dimensions: content overlap, fluency, and structural similarity.

The experiments proved that parameter optimization significantly impacts the efficiency of the model. Changes in settings brought variant scores in the metrics on BLEU and ROUGE, which means finding an optimal balance in parameter adjustment is important to achieve peak performance in Ayurvedic question-answering tasks. The highest score scored for ROUGE-1 precision and BLEU-1 measures indicates the perfection of this model in generating responses relevant and highly similar to the reference text in Ayurveda.

Notable among these results are a ROUGE-1 precision of 0.7051 and a BLEU-1 score of 0.4801, which reflects the model's ability to recover accurate information. These results suggest that the framework could generate outputs that would be more likely to cohere with the seminal Ayurvedic texts, so there is a natural guarantee that these outputs would be high-quality, relevant responses in the traditional Indian medicinal domain. The complete results are provided in Table III. The best configuration obtained is max_new_tokens: 100, temperature=0.6, repetition_penalty=1.2, and top_p=0.9, which forms the strong ground for balanced, relevant, and varied responses.

Advanced hardware such as the Nvidia A100 GPU, along with the implementation of 4-bit quantization, resulted in efficient processing and inference times and made the model

suitable for real application in Ayurvedic clinics, research organizations, as well as education environments. As the system continues to develop and expand, it may play a vital role in the facilitation of Ayurvedic education, research, and practice toward eventually helping further the recognition and integration of Ayurvedic principles into global healthcare systems. Recent advancements in ensemble learning for medical diagnosis [21] and advancements in domain-specific quality assessment [22], highlight the significance of combining multi-model outputs for robust predictions and provide supports applicability of specialized machine learning techniques for better outcomes [23]. Combining images and text data for robust predictions would make significant contributions in this field.

IV. CONCLUSION

The proposed question-answering system, based on Ayurveda, bears numerous advantages in the sphere of traditional medicine relating to crucial questions about the vast amount of historical and contemporary Ayurvedic knowledge. This framework enhances the retrieval and generation of Ayurvedic information much more efficiently and accurately with the help of the combination of Large Language Models together with retrieval-augmented generation methodologies and a vector database. This system permits rapid and accurate retrieval, wherever speedy retrieval becomes a condition of necessity for consultations or scholarly investigations throughout Ayurvedic texts, academic literature, and clinical protocols. This would allow Ayurveda practitioners and researchers to access the most pertinent information instantly, reducing the time and effort that are needed while manually searching through vast datasets, leading further to more prompt decision-making and treatment plans. The proposed framework thus has enormous potential for linguistically accurate and contextually relevant responses by using advanced natural language processing techniques designed specifically to address the unique nuances of Ayurvedic terminology and concepts.

Still, a lot of scope for future improvements and extensions is available. The inclusion of the dataset with a wider range of complex Ayurvedic text may help develop the ability of the model to handle a wider range of linguistic subtleties and contexts specific to different branches of Ayurveda. Another promising path includes multi-lingual features that would make the framework produce answers in numerous Indian languages, this would extend its relevance to areas where Ayurveda is used

TABLE III. PERFORMANCE COMPARISON OF SOME BEST PERFORMING PARAMETER CONFIGURATIONS

Experimental Configurations	Metrics	Rouge 1	Rouge 2	Rouge L	BLEU
Max_new_token = 100; Temp = 0.6; repetition_penalty = 1.2; top_p = 0.9	Precision	0.7051	0.5893	0.6546	0.4801
	Recall	0.5467	0.4562	0.5074	
	F1-score	0.6076	0.5073	0.5642	
Max_new_token = 80; Temp = 0.5; repetition_penalty = 1.1; top_p = 0.95	Precision	0.553	0.3356	0.4535	0.2462
	Recall	0.3472	0.2096	0.2856	
	F1-score	0.4186	0.2523	0.3434	
Max_new_token = 120; Temp = 0.6; repetition_penalty = 1.05; top_p = 0.95	Precision	0.4305	0.2231	0.3226	0.3071
	Recall	0.4091	0.2152	0.3095	
	F1-score	0.4129	0.215	0.3105	

in plenty of linguistic variations. Further research into semantic analysis methodology may enhance the system's abilities to comprehend and render more complex answers within the scheme, particularly in the deciphering of the more complex elements associated with Ayurvedic philosophies and practices.

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Appendix: Sample Queries and Responses from the LLM with RAG after Training related to Ayurveda

S.No.	Queries	Responses
1.	What are some general dietary and lifestyle recommendations (Pathya-Apathya) given in Ayurveda for maintaining eye health?	Some general dietary and lifestyle recommendations (Pathya-Apathya) given in Ayurveda for maintaining eye health include avoiding excessive consumption of black gram and horse gram pulses, sprouts, chilies and spices, sour and fermented foods, heavy foods, junk foods, excess water intake, excess alcohol intake, betel leaf chewing, and certain food combinations that decrease Ojas.
2.	What are some common Ayurvedic treatments for digestive disorders like Amlapitta (acid reflux) ?	Some common Ayurvedic treatments for digestive disorders like Amlapitta (acid reflux) or Atisara (diarrhea) include internal usage of herbal mixtures, external application of oils, and diet and lifestyle modifications. The specific treatment depends on the patient's constitution and symptoms.
3.	How does Ayurveda approach the management of respiratory conditions like Tamaka Shwasa (bronchial asthma)?	Ayurveda approaches the management of respiratory conditions like Tamaka Shwasa through various measures such as Snehana, Snigdha swedana, Parisheka, Dhoomapana, and Panchakarma procedures based on the subtypes of Kaphadhika variant and the patient's tolerance. Additionally, specific herbs and pastes are used for local application and internal administration. The line of treatment involves tailoring the approach according to the severity and chronicity of the condition.
4.	How does Ayurveda describe and treat urinary disorders like Ashmari (urinary calculus) ?	Ayurveda describes the formation of Ashma (stones) within the urinary system as Ashmari, which falls under the category of Mootravaha Srotas Roga and provides detailed descriptions of the various presentations based on doshic predominance, as well as specific treatments and recommendations. These include herbal preparations, dietary advice, and traditional Ayurvedic therapies such as basti and vamana.